

Full Stack Development with MERN

1.Introduction

Project Title

ShopEZ – Online Shopping Platform

Developer

Name	Role	Responsibilities
Ch.Sai Shruthi	Full Stack MERN Developer	Designed and developed the complete ShopEZ application using the MERN Stack. Developed the React.js frontend, implemented RESTful APIs using Node.js and Express.js, designed the MongoDB database, integrated JWT authentication, managed product, cart, and order modules, developed the admin dashboard, performed testing, debugging, deployment, and prepared the complete project documentation.

2. Project Overview

Purpose

ShopEZ is a Full Stack MERN-based Online Shopping Platform designed to provide customers with a secure, reliable, and user-friendly shopping experience. The platform enables users to browse products, search items, manage shopping carts, place orders, and track purchases through an intuitive web interface. Administrators can efficiently manage products, categories, users, banners, and orders through a dedicated dashboard.

Objectives

- Build a complete MERN Stack application.
- Provide secure user authentication using JWT.

- Allow users to browse and search products.
- Enable shopping cart management.
- Support secure checkout and order placement.
- Provide an admin dashboard for managing products, users, and orders.
- Design a scalable MVC-based backend architecture.

Features

Customer Features

- User Registration
- Secure Login
- JWT Authentication
- Product Listing
- Product Search
- Product Filtering
- Product Details
- Shopping Cart
- Checkout
- Order Confirmation
- Order Tracking
- Order History
- Profile Management

Administrator Features

- Admin Login
- Dashboard
- Product Management
- Category Management
- User Management

- Order Management
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3. Architecture

Frontend Architecture

The frontend of ShopEZ – Online Shopping Platform is developed using React.js, HTML, CSS, and JavaScript. It follows a component-based architecture that provides a responsive and user-friendly shopping experience. The frontend enables users to browse products, manage carts, place orders, and track purchases efficiently.

Frontend Modules

- Home Page
 - User Registration
 - User Login
 - Product Catalog
 - Product Details
 - Search & Filter Products
 - Shopping Cart
 - Checkout & Payment
 - Order Tracking
 - User Profile
 - Customer Reviews & Ratings
 - Admin Dashboard
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Backend Architecture

The backend of ShopEZ is developed using Node.js and Express.js. It provides RESTful APIs for user authentication, product management, cart operations, order processing, and admin functionalities. The backend communicates securely with MongoDB and handles all business logic of the application.

Backend Modules

- Authentication API
- User Management API
- Product Management API
- Cart Management API
- Order Management API
- Payment Processing API
- Review & Rating API
- Admin Dashboard API

Routes

REST APIs handle communication between frontend and database.

Database Architecture

ShopEZ uses MongoDB as the database. It stores and manages user details, product information, shopping cart data, order history, reviews, and admin records. MongoDB provides flexibility, scalability, and high performance for e-commerce operations.

Database Collections

- Users
 - Products
 - Categories
 - Cart
 - Orders
 - Reviews
 - Admin
-

4. Setup Instructions

PrerequisitesSystem

Requirements

- Operating System: Windows 10/11 or Linux
- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Internet Connection

Software Requirements

- Node.js (v18 or above)
- MongoDB Atlas / MongoDB Community Edition
- Visual Studio Code
- Git
- Google Chrome Browser

Environment Variables

Create a **.env** file.

PORT=5000

MONGO_URI=your_mongodb_connection

JWT_SECRET=your_secret_key

Prerequisites

- Node.js v18+
- MongoDB (local `mongodb://localhost:27017` or Atlas URI)

1. Clone & Setup Server

```
cd server
npm install
# Edit .env if needed (default: mongodb://localhost:27017/shopez)
npm run seed      # Seeds 16 products + admin + demo user
npm run dev       # Starts server on http://localhost:5000
```

2. Setup Client (new terminal)

```
cd client
npm install
npm run dev      # Starts React app on http://localhost:5173
```

5. Folder Structure

ShopEZ/

```
├── client/           # React Frontend (Vite)
|   ├── public/
|   |   └── _redirects  # Render SPA routing fix
|   └── src/
|       ├── api/        # Axios API calls
|       ├── components/ # Navbar, Footer, ProductCard, etc.
|       ├── context/    # AuthContext, CartContext
|       └── pages/
|           ├── Home.jsx
|           ├── Products.jsx
|           ├── ProductDetail.jsx
|           ├── Cart.jsx
|           ├── Checkout.jsx
|           ├── OrderConfirmation.jsx
|           ├── Profile.jsx
|           ├── Login.jsx
|           ├── Register.jsx
|           └── admin/
|               ├── AdminDashboard.jsx
|               ├── ManageProducts.jsx
|               ├── ManageOrders.jsx
|               └── ManageUsers.jsx
|
└── server/           # Node.js + Express Backend
    ├── config/db.js   # MongoDB connection
```

- └─ middleware/auth.js # JWT middleware
- └─ models/ # User, Product, Cart, Order
- └─ controllers/ # Business logic
- └─ routes/ # API routes
- └─ server.js # Entry point (auto-seeds DB)
- └─ seed.js # Manual DB seeder

6. Running the Application

Backend

cd server

npm install

npm start

Backend runs at:

http://localhost:5000

Frontend

cd client

npm install

npm run dev

Frontend runs at:

http://localhost:5173

7. API Documentation

Authentication APIs

Method	Endpoint	Description
POST	/api/auth/register	Register User

POST /api/auth/login Login User

Product APIs

Method Endpoint

GET /api/products

GET /api/products/

POST /api/products

PUT /api/products/

DELETE /api/products/

Cart APIs

Method Endpoint

GET /api/cart

POST /api/cart/add

PUT /api/cart/update/

DELETE /api/cart/remove/

Order APIs

Method Endpoint

POST /api/orders

GET /api/orders

GET /api/orders/

PUT /api/orders/status/

User APIs

Method Endpoint

GET /api/users/profile

PUT /api/users/profile

API Endpoints

Method	Endpoint	Description	Auth
POST	/api/auth/register	Register new user	Public
POST	/api/auth/login	Login	Public
GET	/api/products	Get all products	Public
GET	/api/products/featured	Featured products	Public
GET	/api/products/:id	Product detail	Public
POST	/api/products/:id/reviews	Add review	User
GET	/api/cart	Get user cart	User
POST	/api/cart/add	Add to cart	User
POST	/api/orders	Place order	User
GET	/api/orders/my	My orders	User
GET	/api/users/profile	Get profile	User
GET	/api/admin/stats	Dashboard stats	Admin
GET	/api/admin/orders	All orders	Admin
PUT	/api/admin/orders/:id/status	Update order status	Admin

8. Authentication

ShopEZ uses JWT (JSON Web Token) authentication.

Authentication flow:

1. User registers.
2. Password is encrypted using bcrypt.
3. User logs in.
4. Server generates JWT Token.

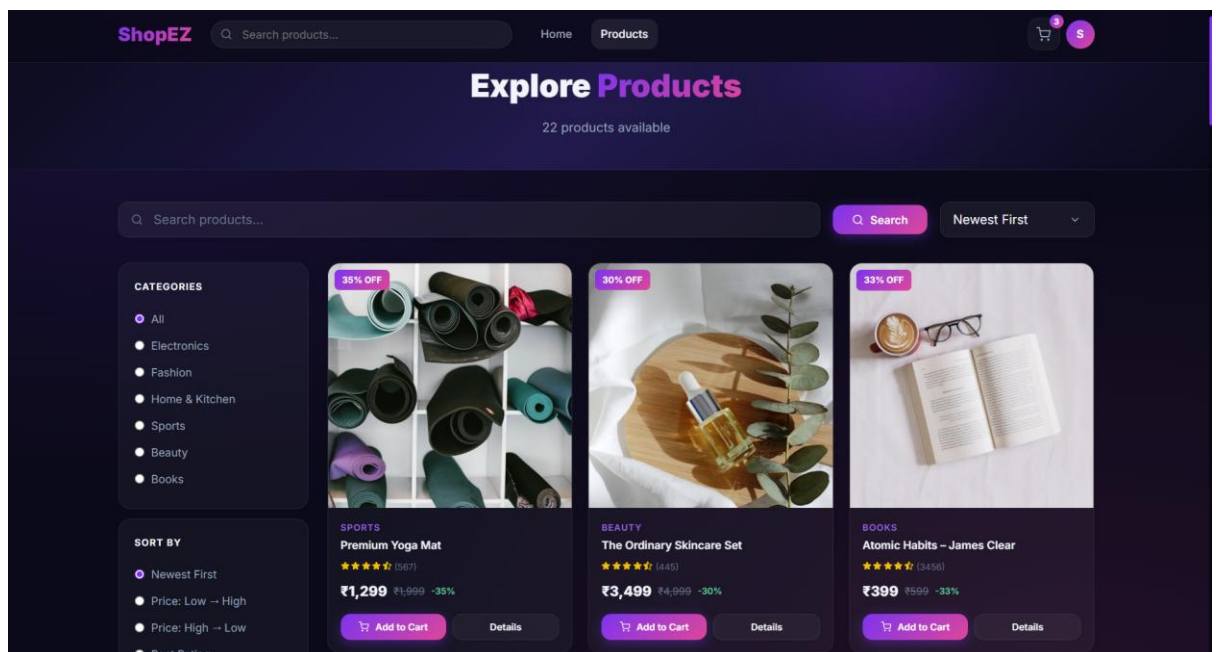
5. Token is stored on the client.
6. Protected routes validate the JWT before granting access.

Role-based authorization separates Customer and Admin privileges.

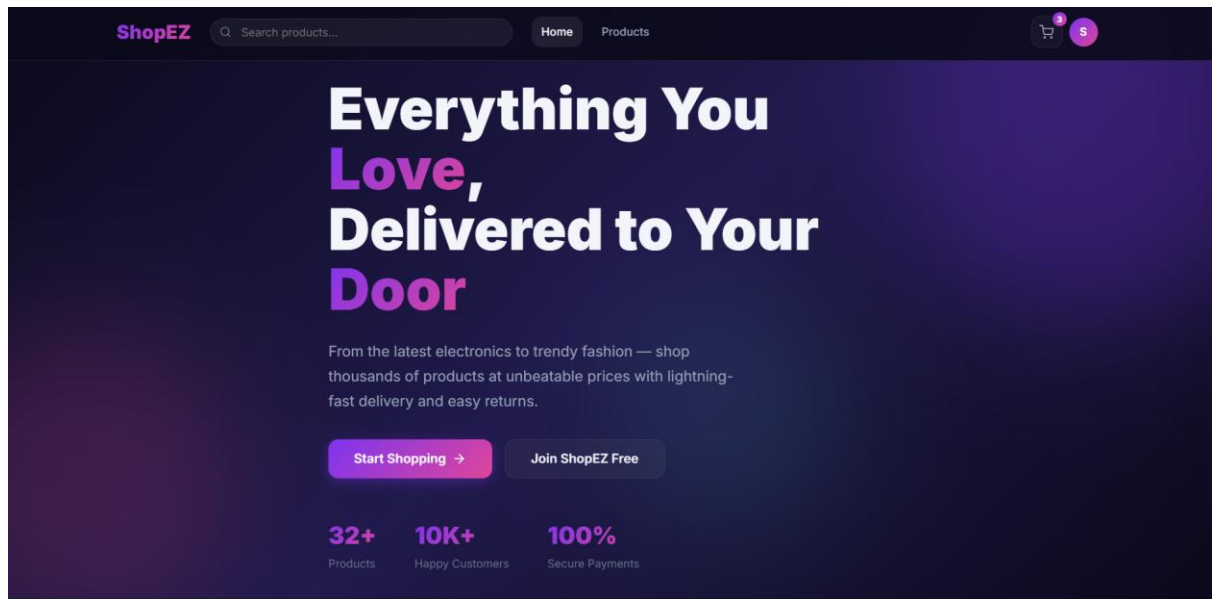
9. User Interface

The application consists of responsive pages:

- Home Page
- Login Page
- Registration Page.
- Shopping Cart
- Checkout
- Orders
- User Profile
- Products Page



Home Page



10. Testing

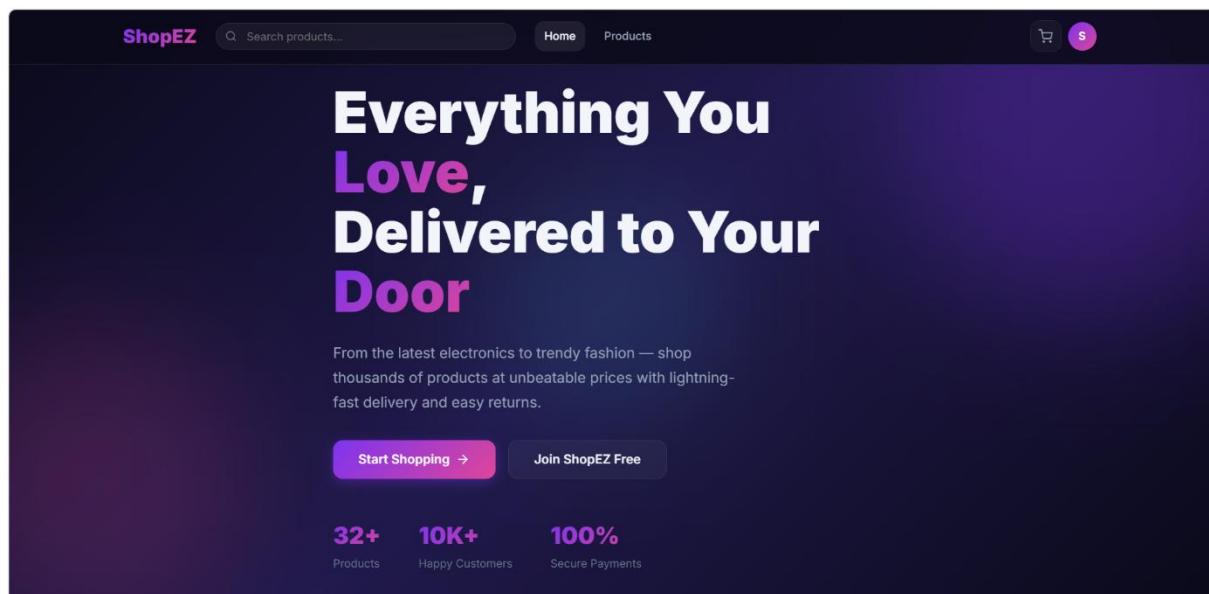
Testing Activities Performed

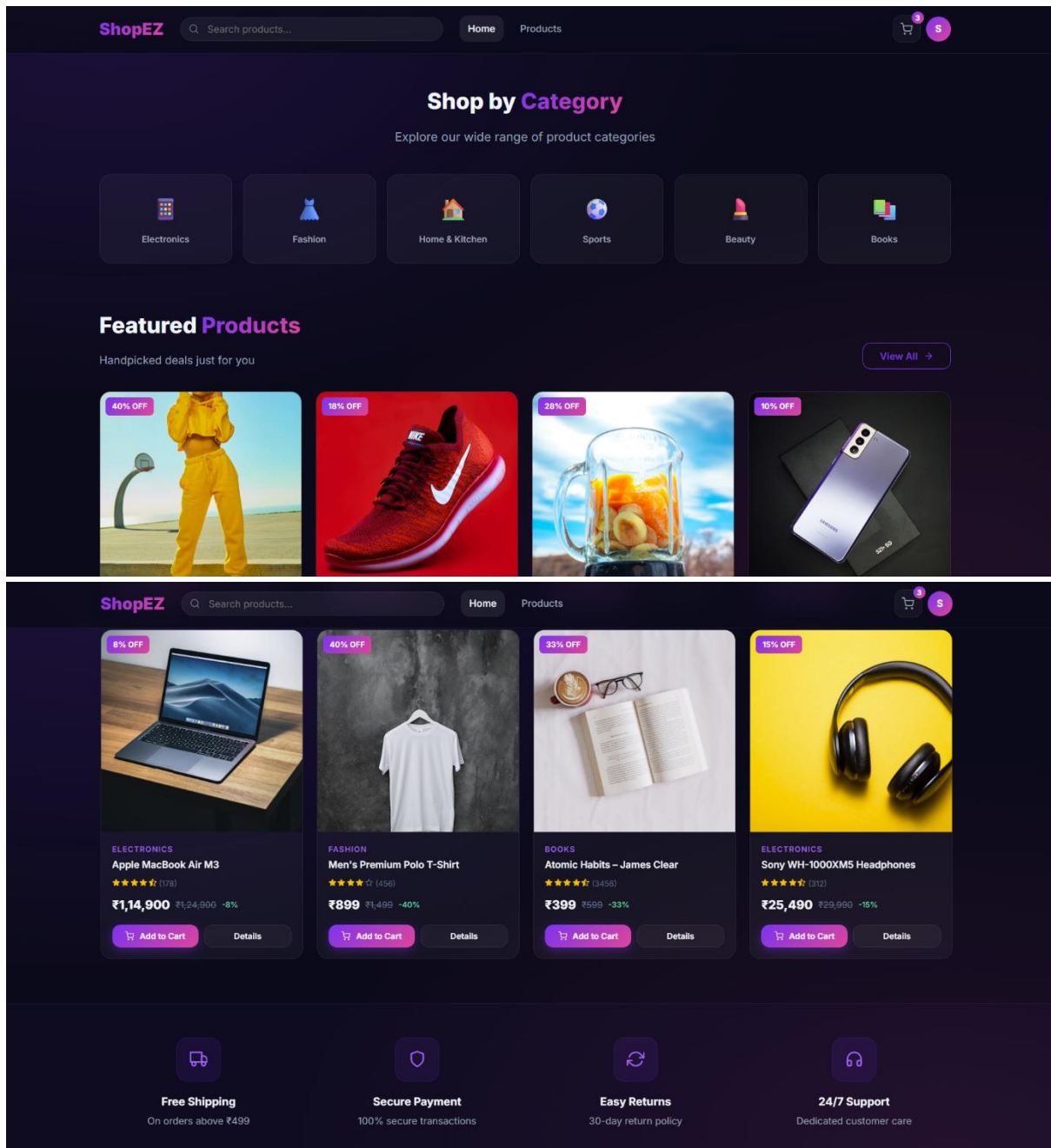
- User Account Registration Validation
- Secure User Login Verification
- Product Search and Filter Testing
- Product Catalog Display Testing
- Shopping Cart Functionality Testing
- Order Placement and Checkout Testing
- Order Tracking Verification
- Customer Profile Management Testing
- Admin Dashboard Functionality Testing
- Product Management Testing
- Order Management Testing
- Database Integration Testing
- API Endpoint Testing
- Security and Session Management Testing

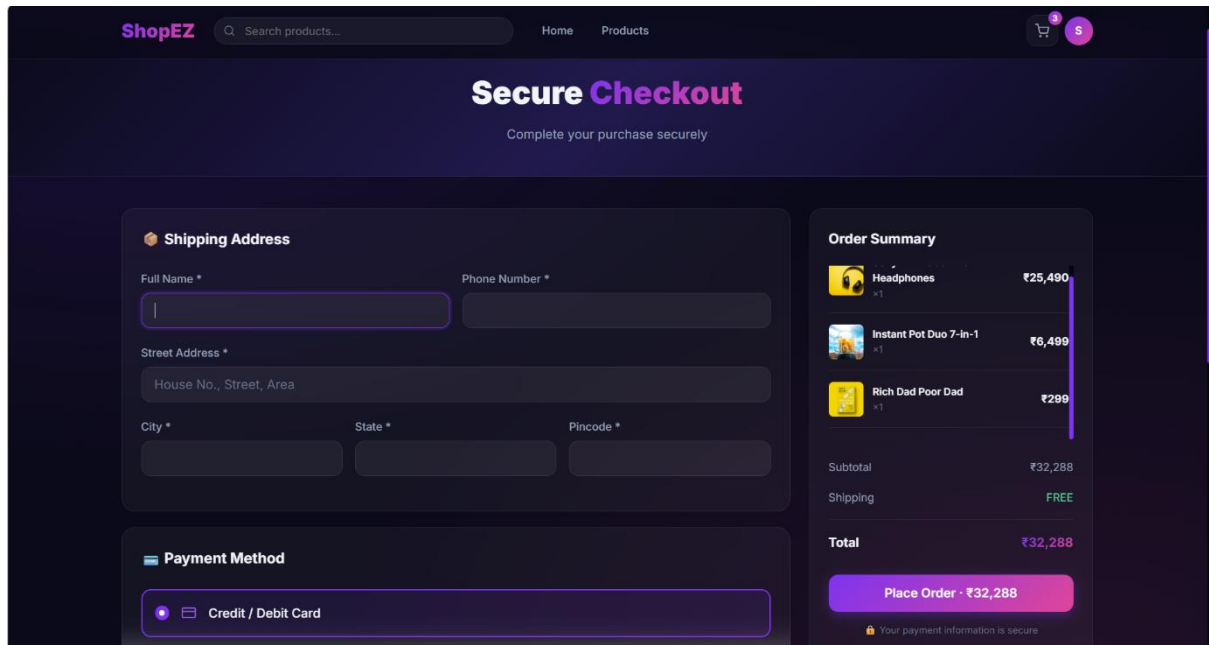
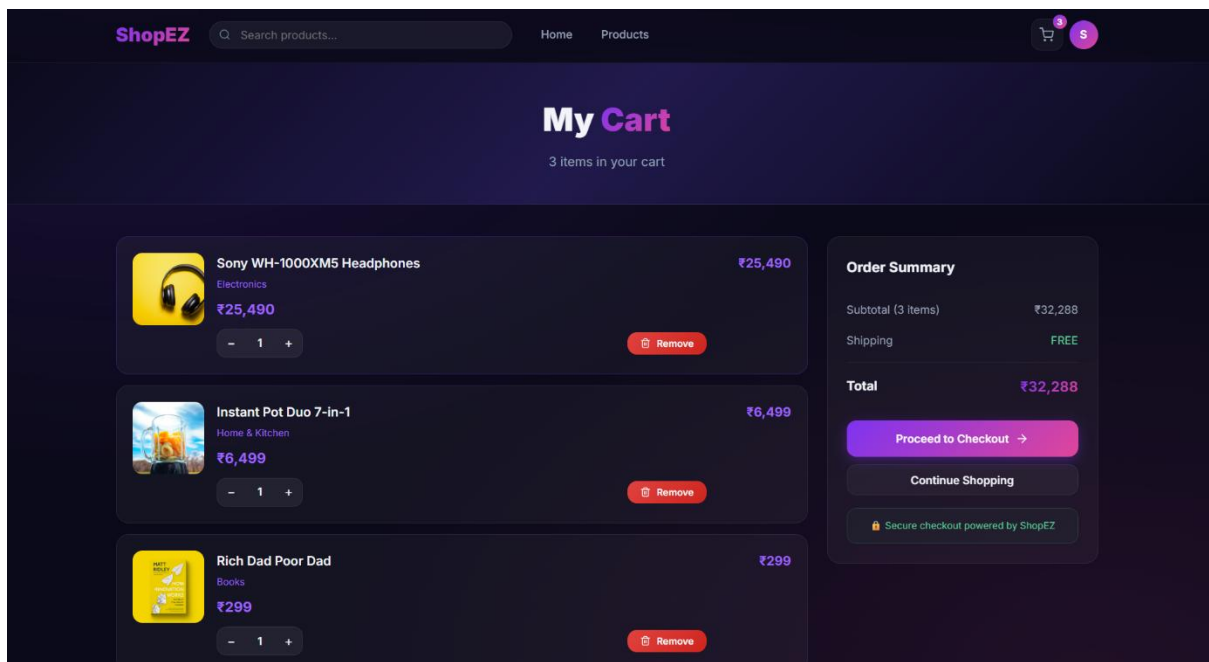
Test Scenarios Covered

- Valid User Registration
- Invalid Login Credentials
- Product Search Accuracy
- Cart Item Addition and Removal
- Successful Order Placement
- Admin Product Updates
- Order Status Updates
- Database Record Validation

11. Screenshots







12. Known Issues

Replace with:

- Product search performance may decrease when the product catalog becomes very large.
- Order status updates may experience slight delays during heavy traffic.

- Admin analytics reports provide only basic sales statistics.
 - Product image uploads support limited file formats.
 - Email notifications may be delayed depending on mail server availability.
 - Cart data is cleared when users manually clear browser storage.
 - Payment gateway integration is currently configured for testing purposes.
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13. Future Enhancements

Replace with:

- Personalized Product Recommendation System
- Advanced Search with Filters and Sorting
- Wishlist and Favorites Module
- Product Reviews and Customer Ratings
- Multiple Payment Options (UPI, Cards, Wallets)
- Discount Coupons and Promotional Offers
- Inventory and Stock Management
- Real-Time Order Tracking
- Mobile Application Development
- Push Notification Service
- Customer Support Chatbot
- Multi-Vendor Marketplace Support
- Multi-Language Support
- Dark Mode Interface
- Sales Analytics and Reporting Dashboard
- Cloud Deployment and Auto Scaling
- AI-Based Customer Behavior Analysis

ShopEZ is a MERN-stack based online shopping platform that provides a secure and user-friendly shopping experience. The system enables customers to browse products, manage carts, place orders, and track purchases efficiently while providing administrators with

powerful product and order management capabilities. Future enhancements will further improve scalability, performance, and customer satisfaction.